STT 843: Multivariate Analysis

10. Discrimination and Classification (Chapter 11.1-11.5)

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Outline

1. Separation and Classification for Two populations
2. Classification with Two Multivariate Normal Populations
3. Fisher's Approach to Classification with Two populations
4. An Allocation Rule Based on Fisher's Discriminant Function

Discrimination and Classification

Discrimination and classification are multivariate techniques concerned with separating distinct sets of objects (or observations) and with allocating new objects (observations) to previously defined group.

- Goal 1. To describe, either graphically (in three or fewer dimensions) or algebraically, the differential features of objects (observations) from several known collections (population). We try to find discriminants whose numerical values are such that the collections are separated as much as possible.
- Goal 2. To sort objects (observations) into two or more labeled classes. The emphasis is on deriving a rule that can be used to optimally assign new objects to the labeled classes.

Separation and Classification for Two populations

Populations π_1 and π_2	Measured variables $\mathbf{X}$
1. Solvent and distressed property-liability insurance companies.	Total assets, cost of stocks and bonds, market value of stocks and bonds, loss expenses, surplus, amount of premiums written.
2. Nonulcer dyspeptics (those with upset stomach problems) and controls ("normal").	Measures of anxiety, dependence, guilt, perfectionism.
3. Federalist Papers written by James Madison and those written by Alexander Hamilton.	Frequencies of different words and lengths of sentences.
4. Two species of chickweed.	Sepal and petal length, petal cleft depth, bract length, scarious tip length, pollen diameter.
5. Purchasers of a new product and laggards (those "slow" to purchase).	Education, income, family size, amount of previous brand switching.
6. Successful or unsuccessful (fail to graduate) college students.	Entrance examination scores, high school grade-point average, number of high school activities.
7. Males and females.	Anthropological measurements, like circumference and volume on ancient skulls.
8. Good and poor credit risks.	Income, age, number of credit cards, family size.
9. Alcoholics and nonalcoholics.	Activity of monoamine oxidase enzyme, activity. of adenylate cyclase enzyme.

- Allocation or classification rules are usually developed from learning samples. Measured characteristics of randomly selected object known to come from each of the two populations are examined for differences.
- Why we know that some observations belong to a particular population, but we are unsure about others.
  - Incomplete knowledge of future performance
  - Perfect information requires destroying the object.
  - Unavailable or expensive information.

Example (Discriminating owners from nonowners of riding mowers)

- Consider two groups in a city: π_1, riding-mower owners, and π_2, those without riding mowersthat is, nonowners.
- In order to identify the best sales prospects for an intensive sales campaign, a riding-mower manufacturer is interested in classifying families as prospective owners or nonowners on the basis of $x_1 =$ income and $x_2 =$ lot size.
- Random samples of $n_1 = 12$ current owners and $n_2 = 12$ current nonowners yield the values in the following table

π_1 : Riding-mower owners		π_2 : Nonowners	
x_1 (Income in \$1000 s)	x_2 (Lot size in 1000ft^2)	x_1 (Income 1000 s)	x_2 (Lot size in 1000ft^2)
90.0	18.4	105.0	19.6
115.5	16.8	82.8	20.8
94.8	21.6	94.8	17.2
91.5	20.8	73.2	20.4
117.0	23.6	114.0	17.6
140.1	19.2	79.2	17.6
138.0	17.6	89.4	16.0
112.8	22.4	96.0	18.4
99.0	20.0	77.4	16.4
123.0	20.8	63.0	18.8
81.0	22.0	81.0	14.0
111.0	20.0	93.0	14.8

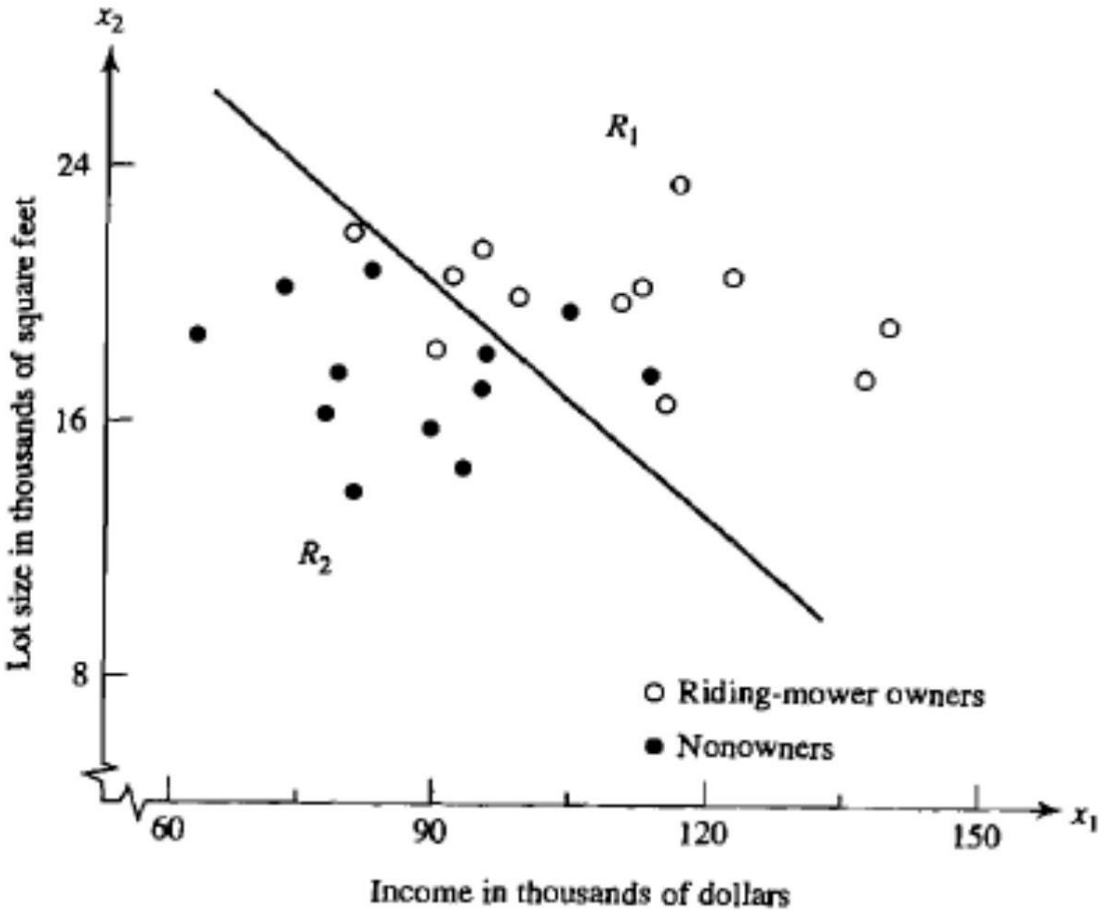


Figure: Income and lot size for riding-mower owners and nonowners.

Classification regions

- Let $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ be the probability density functions associated with $p \times 1$ vector random variable $\boldsymbol{X}$ for the populations π_1 and π_2 respectively.
- An object with associate measurements $\mathbf{x}$ must be assigned to either π_1 or π_2.
- Let Ω be the complete sample space, let R_1 be that set of $\mathbf{x}$ values for which we classify objects as π_1 and $R_2 = \Omega - R_1$ be the remaining $\mathbf{x}$ values for which we classify objects as π_2. So $R_1 \cup R_2 = \Omega$ and $R_1 \cap R_2 = \emptyset$.

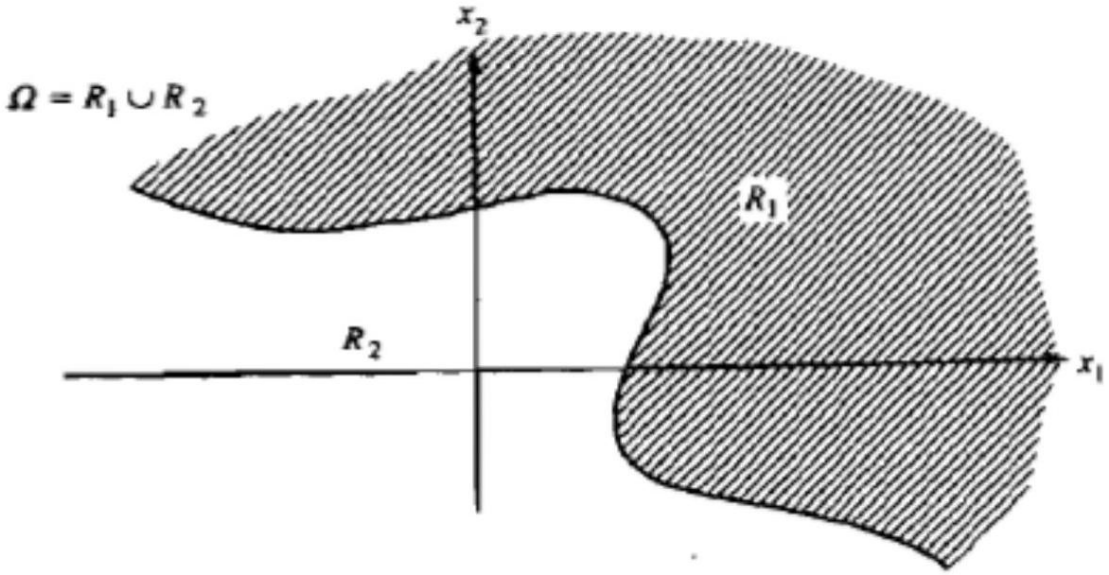


Figure: Classification regions for two populations.

Make decisions

- The conditional probability, $P(2 \mid 1)$, of classifying an object as π_2 when, in fact, it is from π_1 is

$$P(2 \mid 1) = P\left(\boldsymbol{X} \in R_2 \mid \pi_1\right) = \int_{R_2 = \Omega - R_1} f_1(\mathbf{x}) d\mathbf{x}$$

- Similar, the conditional probability, $P(1 \mid 2)$, of classifying an object as π_1 when, in fact, it is from π_2 is

$$P(1 \mid 2) = P\left(\boldsymbol{X} \in R_1 \mid \pi_2\right) = \int_{R_1 = \Omega - R_2} f_1(\mathbf{x}) d\mathbf{x}$$

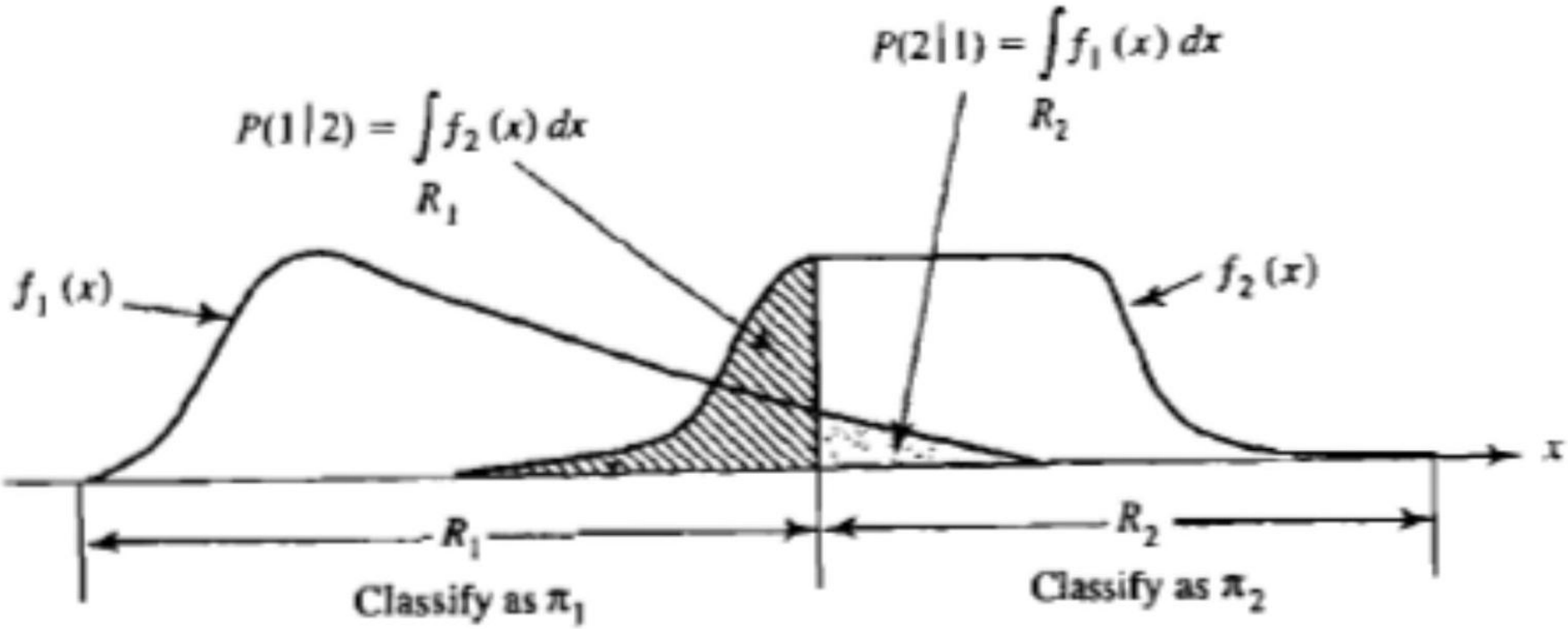


Figure: Misclassification probabilities for hypothetical classification regions when $p = 1$.

Let p_1 be the prior probability of π_1 and p_2 be the prior probability of π_2, where $p_1 + p_2 = 1$. Then

$$
\begin{aligned}
& P\left(\text{ observation is correctly classified as } \pi_1\right) \\
=& P\left(\boldsymbol{X} \in R_1 \mid \pi_1\right) P\left(\pi_1\right) = P(1 \mid 1) p_1, \\
& P\left(\text{ observation is misclassified as } \pi_1\right) \\
=& P\left(\boldsymbol{X} \in R_1 \mid \pi_2\right) P\left(\pi_2\right) = P(1 \mid 2) p_2, \\
& P\left(\text{ observation is correlectly classified as } \pi_2\right) \\
=& P\left(\boldsymbol{X} \in R_2 \mid \pi_2\right) P\left(\pi_2\right) = P(2 \mid 2) p_2, \\
& P\left(\text{ observation is misclassified as } \pi_2\right) \\
=& P\left(\boldsymbol{X} \in R_2 \mid \pi_1\right) P\left(\pi_1\right) = P(2 \mid 1) p_1,
\end{aligned}
$$

- The costs of misclassification is defined by a cost matrix

	π_1	π_2
π_1	0	$c(2 \mid 1)$
π_2	$c(1 \mid 2)$	0

- Expected cost of misclassification (ECM)

$$\mathrm{ECM} = c(2 \mid 1)P(2 \mid 1)p_1 + c(1 \mid 2)P(1 \mid 2)p_2$$

A reasonable classification rule should have an ECM as small, or nearly as small, as possible.

Results 11.1

The region R_1 and R_2 that minimize the ECM are defined by the value $\mathbf{x}$ for which the following inequalities hold:

$$R_1 : \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \left(\frac{c(1 \mid 2)}{c(2 \mid 1)}\right)\left(\frac{p_2}{p_1}\right)$$

$$\begin{pmatrix} \text{density} \\ \text{ratio} \end{pmatrix} \geq \begin{pmatrix} \text{cost} \\ \text{ratio} \end{pmatrix} \begin{pmatrix} \text{prior probability} \\ \text{ratio} \end{pmatrix}$$

$$R_2 : \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \left(\frac{c(1 \mid 2)}{c(2 \mid 1)}\right)\left(\frac{p_2}{p_1}\right)$$

$$\begin{pmatrix} \text{density} \\ \text{ratio} \end{pmatrix} < \begin{pmatrix} \text{cost} \\ \text{ratio} \end{pmatrix} \begin{pmatrix} \text{prior probability} \\ \text{ratio} \end{pmatrix}$$

Special Cases of Minimum Expected Cost Regions

(a) $p_2/p_1 = 1$ (equal prior probabilities)

$$R_1 : \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{c(1 \mid 2)}{c(2 \mid 1)} \quad R_2 : \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{c(1 \mid 2)}{c(2 \mid 1)}$$

(b) $c(1 \mid 2)/c(2 \mid 1) = 1$ (equal misclassification costs)

$$R_1 : \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{p_2}{p_1} \quad R_2 : \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{p_2}{p_1}$$

(c) $p_2/p_1 = c(1 \mid 2)/c(2 \mid 1) = 1$ or $p_2/p_1 = 1/(c(1 \mid 2)/c(2 \mid 1))$
(equal prior probabilities and equal misclassification costs)

$$R_1 : \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq 1 \quad R_2 : \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < 1$$

Example (Classifying a new observation into one of the two populations)

- A researcher has enough data available to estimate the density functions $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ associated with populations π_1 and π_2, respectively. Suppose $c(2 \mid 1) = 5$ units and $c(1/2) = 10$ units.
- In addition, it is known that about 20% of all objects (for which the measurements $\mathbf{x}$ can be recorded) belong to π_2. Thus, the prior probabilities are $p_1 = .8$ and $p_2 = .2$.

Given the prior probabilities and costs of misclassification, we can use (11-6) to derive the classification regions R_1 and R_2. Specifically, we have

$$R_1 : \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \left(\frac{10}{5}\right)\left(\frac{.2}{.8}\right) = .5$$
$$R_2 : \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \left(\frac{10}{5}\right)\left(\frac{.2}{.8}\right) = .5$$

Suppose the density functions evaluated at a new observation $\mathbf{x}_0$ give $f_1(\mathbf{x}_0) = .3$ and $f_2(\mathbf{x}_0) = .4$. Do we classify the new observation as π_1 or π_2? To answer the question, we form the ratio

$$\frac{f_1(\mathbf{x}_0)}{f_2(\mathbf{x}_0)} = \frac{.3}{.4} = .75$$

and compare it with .5 obtained before. Since

$$\frac{f_1(\mathbf{x}_0)}{f_2(\mathbf{x}_0)} = .75 > \left(\frac{c(1 \mid 2)}{c(2 \mid 1)}\right)\left(\frac{p_2}{p_1}\right) = .5$$

we find that $\mathbf{x}_0 \in R_1$ and classify it as belonging to π_1.

Other classification procedures:

- Choose R_1 and R_2 to minimize the total probability of misclassification (TPM).

$$\mathrm{TPM} = p_1 \int_{R_2} f_1(\mathbf{x}) d\mathbf{x} + p_2 \int_{R_1} f_2(\mathbf{x}) d\mathbf{x}$$

- Allocate a new observation $\mathbf{x}_0$ to the population with the largest posterior probability $P(\pi_i \mid \mathbf{x}_0)$.

$$P(\pi_1 \mid \mathbf{x}_0) = \frac{p_1 f_1(\mathbf{x}_0)}{p_1 f_1(\mathbf{x}_0) + p_2 f_2(\mathbf{x}_0)}$$

$$P(\pi_2 \mid \mathbf{x}_0) = 1 - P(\pi_1 \mid \mathbf{x}_0) = \frac{p_2 f_2(\mathbf{x}_0)}{p_1 f_1(\mathbf{x}_0) + p_2 f_2(\mathbf{x}_0)}.$$

Classifying an observation $\mathbf{x}_0$ as π_1 when $P(\pi_1 \mid \mathbf{x}_0) > P(\pi_2 \mid \mathbf{x}_0)$ is equivalent to using the (b) rule (Minimum Expected Cost Regions) for total probability of misclassification.

Classification with Two Multivariate Normal Populations

- Assume $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ are multivariate normal densities, the first with mean vector $\boldsymbol{\mu}_1$ and covariance Σ_1 and the second with mean $\boldsymbol{\mu}_2$ and covariance Σ_2.
- Suppose that joint densities of $\boldsymbol{X}^\mathsf{T} = [X_1, X_2, \ldots, X_p]$ for population π_1 and π_2 are given by

$$f_i(\mathbf{x}) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^\mathsf{T}\Sigma(\mathbf{x}-\boldsymbol{\mu}_i)\right],$$

for $i = 1, 2$ and $\Sigma_1 = \Sigma_2 = \Sigma$

Result: When $\Sigma_1 = \Sigma_2 = \Sigma$

Let the populations π_1 and π_2 be described by multivariate normal densities of the form above. Then the allocation rule that minimizes the ECM is as follows:

Allocate $\mathbf{x}_0$ to π_1 if

$$(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^{\mathsf{T}} \Sigma^{-1} \mathbf{x}_0 - \frac{1}{2} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \Sigma^{-1} (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)$$
$$\geq \ln \left[\left(\frac{c(1 \mid 2)}{c(2 \mid 1)} \right) \left(\frac{p_2}{p_1} \right) \right],$$

Allocate $\mathbf{x}_0$ to π_2 otherwise.

The Estimated Minimum ECM Rule for Two Normal Population

Allocate $\mathbf{x}_0$ to π_1 if

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)^\top \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x}_0 - \frac{1}{2}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)\mathbf{S}_{\text{pooled}}^{-1}(\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2)$$
$$\geq \ln\left[\left(\frac{c(1 \mid 2)}{c(2 \mid 1)}\right)\left(\frac{p_2}{p_1}\right)\right],$$

Allocate $\mathbf{x}_0$ to π_2 otherwise.

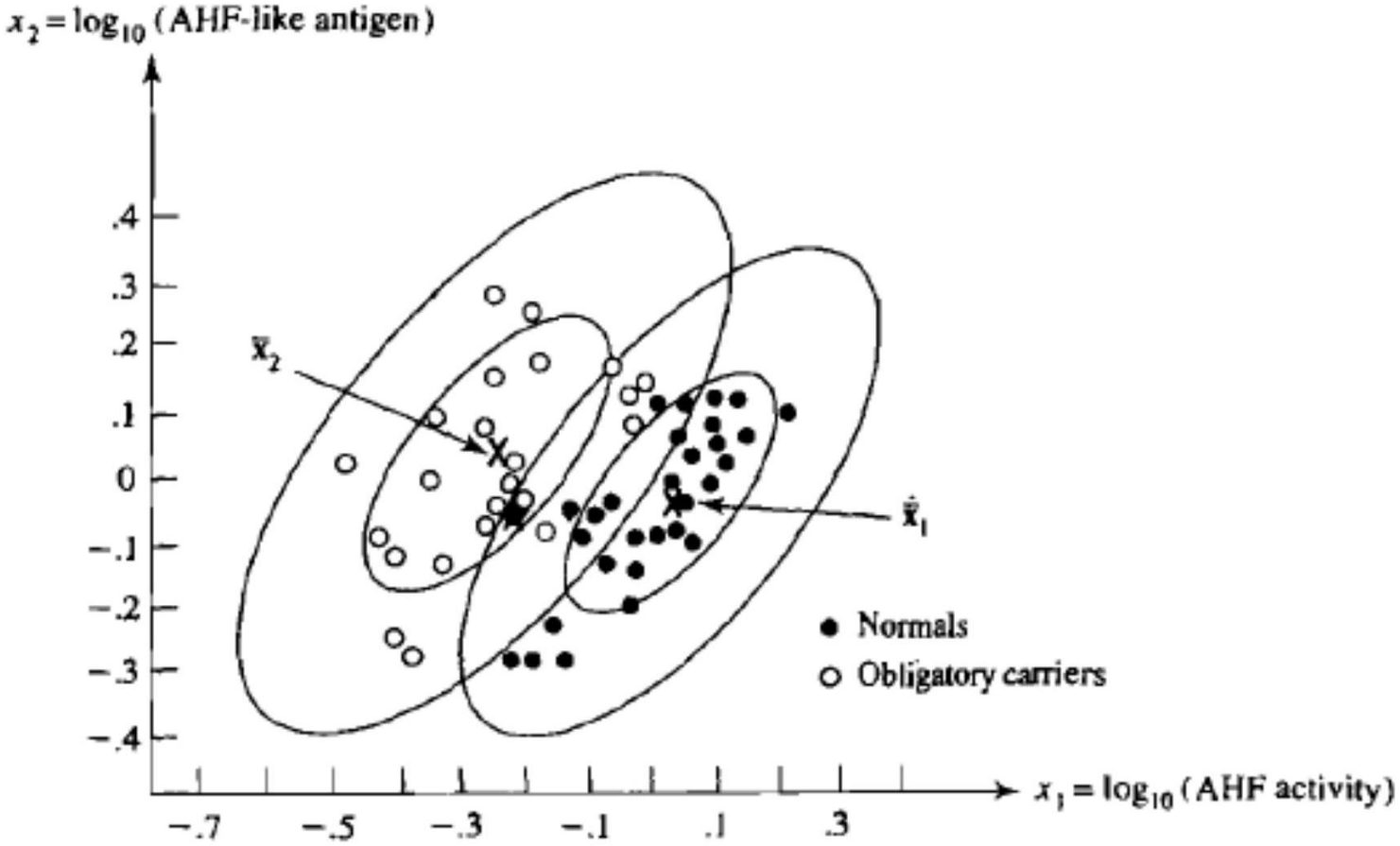
Example 11.3 (Classification with two normal populations-common $\mathbf{\Sigma}$ and equal costs)

- This example is adapted from a study concerned with the detection of hemophilia A carriers.
- To construct a procedure for detecting potential hemophilia A carriers, blood samples were assayed for two groups of women and measurements on the two variables,

$$X_1 = \log_{10}(\text{ AHF activity })$$
$$X_2 = \log_{10}(\text{ AHF-like antigen })$$

recorded. ("AHF" denotes antihemophilic factor.)

- The first group of $n_1 = 30$ women were selected from a population of women who did not carry the hemophilia gene. This group was called the *normal* group.
- The second group of $n_2 = 22$ women was selected from known hemophilia A carriers (daughters of hemophiliacs, mothers with more than one hemophilic son, and mothers with one hemophilic son and other hemophilic relatives). This group was called the *obligatory* carriers.


$x_2 = \log_{10}$ (AHF-like antigen)
.4
.3
.2
.1
0
−.1
−.2
−.3
−.4
$\bar{x}_2$
$\bar{x}_1$
Normals
Obligatory carriers
−.7
−.5
−.3
−.1
.1
.3
$x_1 = \log_{10}$ (AHF activity)

$$\bar{\mathbf{x}}_1 = \begin{bmatrix} -.0065 \\ -.0390 \end{bmatrix}, \quad \bar{\mathbf{x}}_2 = \begin{bmatrix} -.2483 \\ .0262 \end{bmatrix}$$

$$\mathbf{S}_{\text{pooled}}^{-1} = \begin{bmatrix} 131.158 & -90.423 \\ -90.423 & 108.147 \end{bmatrix}$$

Therefore, the equal costs and equal priors discriminant function is

$$
\begin{aligned}
\hat{y} &= \hat{\mathbf{a}}^{\mathsf{T}}\mathbf{x} = [\overline{\mathbf{x}}_1 - \overline{\mathbf{x}}_2]^{\mathsf{T}}\, \mathbf{S}_{\text{pooled}}^{-1}\, \mathbf{x} \\
&= \begin{bmatrix} .2418 & -.0652 \end{bmatrix} \begin{bmatrix} 131.158 & -90.423 \\ -90.423 & 108.147 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
&= 37.61x_1 - 28.92x_2
\end{aligned}
$$

Moreover,

$$\bar{y}_1 = \hat{\mathbf{a}}^{\mathsf{T}}\bar{\mathbf{x}}_1 = \begin{bmatrix} 37.61 & -28.92 \end{bmatrix} \begin{bmatrix} -.0065 \\ -.0390 \end{bmatrix} = .88$$

$$\bar{y}_2 = \hat{\mathbf{a}}^{\mathsf{T}}\bar{\mathbf{x}}_2 = \begin{bmatrix} 37.61 & -28.92 \end{bmatrix} \begin{bmatrix} -.2483 \\ .0262 \end{bmatrix} = -10.10$$

and the midpoint between these means is

$$\hat{m} = \frac{1}{2}(\bar{y}_1 + \bar{y}_2) = \frac{1}{2}(.89 - 10.10) = -4.61$$

Measurements of AHF activity and AHF-like antigen on a woman who may be a hemophilia A carrier give $x_1 = -.210$ and $x_2 = -.044$. Should this woman be classified as π_1 (normal) or π_2 (obligatory carrier)?

With equal costs and equal priors, $\ln(1) = 0$, we obtain
Allocate $\mathbf{x}_0$ to π_1 if $\hat{y}_0 = \hat{\mathbf{a}}^\mathsf{T}\mathbf{x}_0 \geq \hat{m} = -4.61$
Allocate $\mathbf{x}_0$ to π_2 if $\hat{y}_0 = \hat{\mathbf{a}}^\mathsf{T}\mathbf{x}_0 < \hat{\boldsymbol{m}} = -4.61$ where
$\mathbf{x}_0^\mathsf{T} = [-.210, -.044]$. Since

$$\hat{y}_0 = \hat{\mathbf{a}}^\mathsf{T}\mathbf{x}_0 = \begin{bmatrix} 37.61 & -28.92 \end{bmatrix} \begin{bmatrix} -.210 \\ -.044 \end{bmatrix} = -6.62 < -4.61$$

we classify the woman as π_2, an obligatory carrier.

- Suppose now that the prior probabilities of group membership are known. For example, the prior probabilities of group membership are $p_1 = .75$ and $p_2 = .25$.
- Assuming, $c(1 \mid 2) = c(2 \mid 1)$, and using the classification statistic

$$\hat{w} = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)^{\mathsf{T}} \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x}_0 - \frac{1}{2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)^{\mathsf{T}} \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2)$$

- Or $\hat{w} = \hat{\mathbf{a}}^{\mathsf{T}} \mathbf{x}_0 - \hat{m}$ with $\mathbf{x}_0^{\mathsf{T}} = [-.210, -.044]$, $\hat{m} = -4.61$, and $\hat{\mathbf{a}}^{\mathsf{T}} \mathbf{x}_0 = -6.62$, we have

$$\hat{w} = -6.62 - (-4.61) = -2.01$$

we see that

$$\hat{w} = -2.01 < \ln\left[\frac{p_2}{p_1}\right] = \ln\left[\frac{.25}{.75}\right] = -1.10$$

and we classify the woman as π_2, an obligatory carrier.

Scaling

- The coefficient vector $\hat{\mathbf{a}} = \mathbf{S}_{\text{pooled}}\ (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$ is unique only up to a multiplicative constant, so for $c \neq 0$, any vector $c\hat{\mathbf{a}}$ will also serve as discriminant coefficients.
- The vector $\hat{a}$ is frequently "scaled" or "normalized" to ease the interpretation of its elements.
- (1) Set $\hat{\mathbf{a}}^* = \hat{\mathbf{a}}/\sqrt{\hat{\mathbf{a}}^\mathsf{T}\hat{\mathbf{a}}}$.
- (2) Set $\hat{\mathbf{a}}^* = \hat{\mathbf{a}}/\hat{a}_1$.

Fisher's Approach to Classification with Two populations

- Fisher's idea was to transform the multivariate observations x to univariate observation y such that y 's derived from populations π_1 and π_2 were separated as much as possible.
- Fisher suggested taking linear combinations of $\mathbf{x}$ to create y 's because they are simple enough functions of the $\mathbf{x}$ to be handled easily.
- Fisher's approach does not assume that the populations are normal, but implicitly assume that the population covariance matrices are equal.

Result 6-2.3

The linear combinations $\hat{y} = \hat{\mathbf{a}}^{\mathsf{T}}\mathbf{x} = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)^{\mathsf{T}} \mathbf{S}_{\text{pooled}}^{-1} \mathbf{x}$ maximize the ratio

$$\frac{\left(\begin{array}{c}\text{squared distance}\\ \text{between sample mean of } y\end{array}\right)}{(\text{ sample variance of } y)} = \frac{(\bar{y}_1 - \bar{y}_2)^2}{s_y^2}$$

$$= \frac{\left(\hat{\mathbf{a}}^{\mathsf{T}}\bar{\mathbf{x}}_1 - \hat{\mathbf{a}}^{\mathsf{T}}\bar{\mathbf{x}}_2\right)^2}{\hat{\mathbf{a}}^{\mathsf{T}}\mathbf{S}_{\text{pooled}}\,\hat{\mathbf{a}}} = \frac{\left(\hat{\mathbf{a}}^{\mathsf{T}}\mathbf{d}\right)^2}{\hat{\mathbf{a}}^{\mathsf{T}}\mathbf{S}_{\text{pooled}}\,\hat{\mathbf{a}}}$$

over all possible coefficient vectors $\hat{\mathbf{a}}$ where $\mathbf{d} = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$. The maximum of the ratio is

$$D^2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)^{\mathsf{T}} \mathbf{S}_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2).$$

Allocate $\mathbf{x}_0$ to π_1 if

$$\hat{y}_0 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)\,\mathbf{S}_{\text{pooled}}^{-1}\,\mathbf{x}_0 \geq \hat{m} = \frac{1}{2}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)\,\mathbf{S}_{\text{pooled}}^{-1}\,(\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2)$$

or

$$\hat{y}_0 - \hat{m} \geq 0.$$

Allocate $\mathbf{x}_0$ to π_2 if

$$\hat{y}_0 < \hat{m} \quad \text{or} \quad \hat{y}_0 - \hat{m} < 0.$$

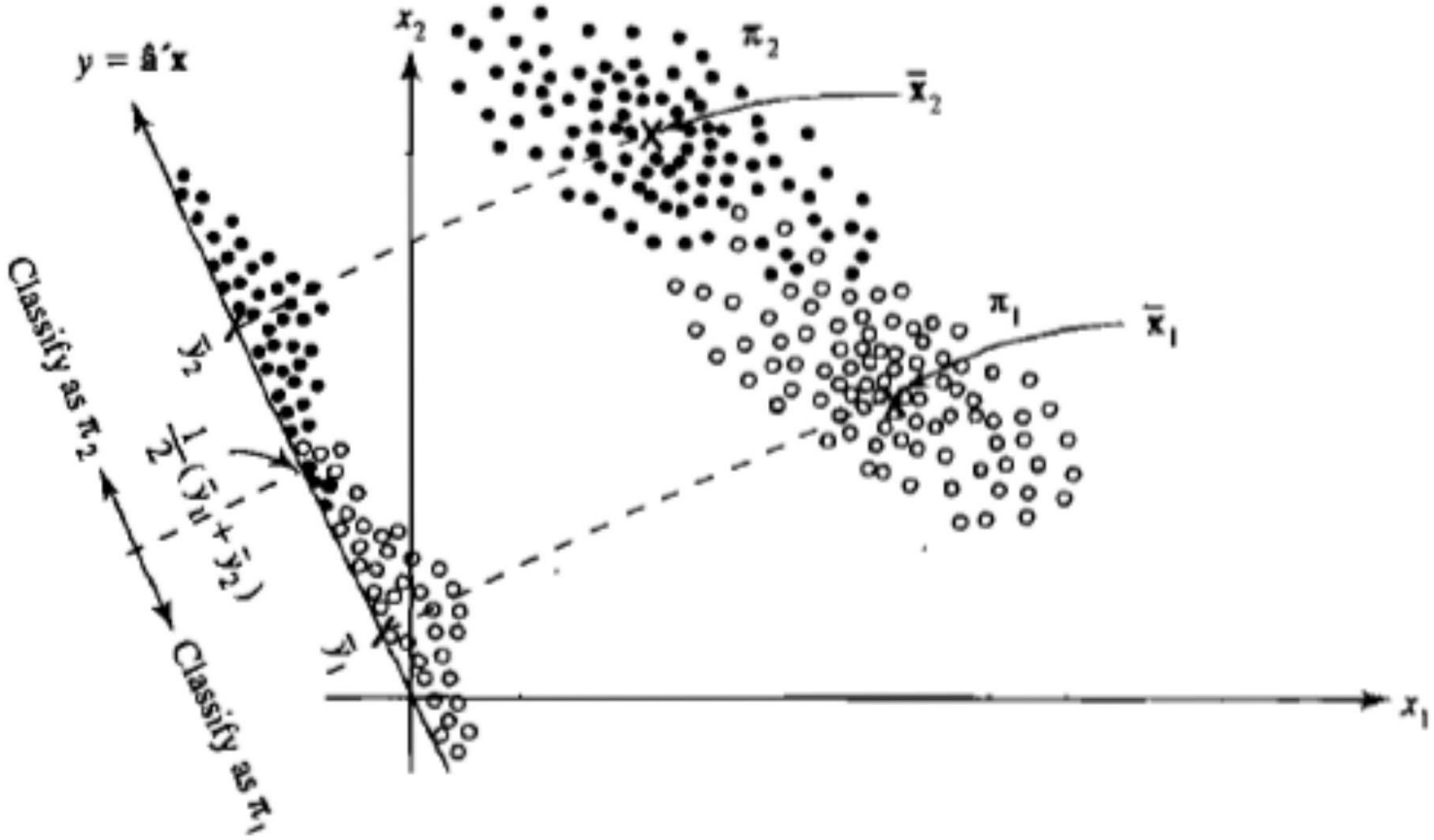


Figure II.5 A pictorial representation of Fisher's procedure for two populations with $p = 2$.

Example II.4 (Fisher's linear discriminant for the hemophilia data)

Consider the detection of hemophilia A carriers introduced in Example 11.3. Recall that the equal costs and equal priors linear discriminant function was

$$\hat{y} = \hat{\mathbf{a}}^{\mathsf{T}}\mathbf{x} = \left(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2\right)^{\mathsf{T}} \mathbf{S}_{\text{pooled}}^{-1}\, \mathbf{x} = 37.61x_1 - 28.92x_2$$

This linear discriminant function is Fisher's linear function, which maximally separates the two populations, and the maximum separation in the samples is

$$\begin{aligned} D^2 &= \left(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2\right)^{\mathsf{T}} \mathbf{S}_{\text{pooled}}^{-1}\, \left(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2\right) \\ &= \begin{bmatrix} .2418, & -.0652 \end{bmatrix} \begin{bmatrix} 131.158 & -90.423 \\ -90.423 & 108.147 \end{bmatrix} \begin{bmatrix} .2418 \\ -.0652 \end{bmatrix} \\ &= 10.98 \end{aligned}$$

Classification of Normal Population When $\Sigma_1 \neq \Sigma_2$

Let the populations π_1 and π_2 be described by multivariate normal densities with mean vectors and covariance matrices $\boldsymbol{\mu}_1, \Sigma_1$ and $\boldsymbol{\mu}_2$, Σ_2, respectively. The allocation rule that minimizes the expected cost of misclassification is given by
Allocate $\mathbf{x}_0$ to π_1 if

$$-\frac{1}{2}\mathbf{x}_0^{\mathsf{T}}\left(\Sigma_1^{-1}-\Sigma_2^{-1}\right)\mathbf{x}_0+\left(\boldsymbol{\mu}_1^{\mathsf{T}}\Sigma_1^{-1}-\boldsymbol{\mu}_2^{\mathsf{T}}\Sigma_2^{-1}\right)\mathbf{x}_0-k$$
$$\geq \ln\left[\left(\frac{c(1\mid 2)}{c(2\mid 1)}\right)\left(\frac{p_2}{p_1}\right)\right],$$

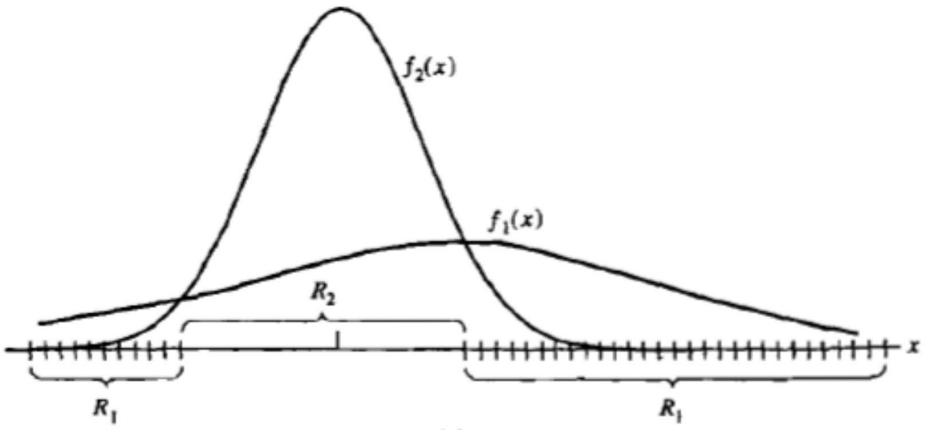
where $k=\frac{1}{2}\ln\left(\frac{|\Sigma_1|}{|\Sigma_2|}\right)+\frac{1}{2}\left(\boldsymbol{\mu}_1^{\mathsf{T}}\Sigma_1^{-1}\boldsymbol{\mu}_1-\boldsymbol{\mu}_2^{\mathsf{T}}\Sigma_2^{-1}\boldsymbol{\mu}_2\right)$. Allocate $\mathbf{x}_0$ to π_2 otherwise

Quadratic Classification Rule (Normal Populations with Unequal Covariance Matrices)

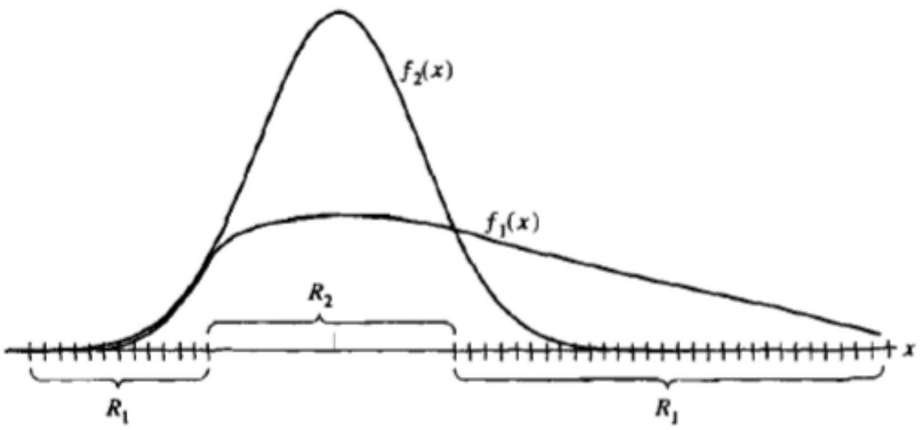
Allocate $\mathbf{x}_0$ to π_1 if

$$-\frac{1}{2}\mathbf{x}_0^{\mathsf{T}}\left(\mathbf{S}_1^{-1}-\mathbf{S}_2^{-1}\right)\mathbf{x}_0+\left(\bar{\mathbf{x}}_1^{\mathsf{T}}\mathbf{S}_1^{-1}-\bar{\mathbf{x}}_2^{\mathsf{T}}\mathbf{S}_2^{-1}\right)\mathbf{x}_0-k$$
$$\geq \ln\left[\left(\frac{c(1\mid 2)}{c(2\mid 1)}\right)\left(\frac{p_2}{p_1}\right)\right]$$

Allocate $\mathbf{x}_0$ to π_2 otherwise.



(a)



(b)

Figure: Quadratic rules for (a) two normal distribution with unequal variances and (b) two distributions, one of which is nonnormal-rule not appropriate.